

Find Alex ARG: Mechanics and Cybersecurity Themes

Joshua Sourbeer
jo501433@ucf.edu
University of Central Florida
Orlando, Florida, USA

Jasmine Darman
jasmine.darman@ucf.edu
University of Central Florida
Orlando, Florida, USA

Jennifer Tyler
jennifer.tyler@ucf.edu
University of Central Florida
Orlando, Florida, USA

Matthew Mosher
matthew.mosher@ucf.edu
University of Central Florida
Orlando, Florida, USA

Abstract

Alternate reality games (ARGs) are often intended to be entertaining and intriguing, blurring the realism of everyday life with the prospect of a more exciting and frequently unattainable experience. Many ARGs involve puzzles or challenges that players complete to progress through an interactive storyline, culminating in the completion of the narrative. By examining the Find Alex ARG, a comprehensive analysis is presented of how the ARG was built to serve as a narratively fascinating mystery game and convey realistic elements themed on cybersecurity. This analysis will uncover how developers were able to manage a player-centric design process while operating within a zero-budget environment. Within each facet of the presented analysis, a detailed examination of the building blocks and mechanical elements of Find Alex is presented from a developer perspective.

CCS Concepts

• **Human-centered computing** → *User interface design; Interface design prototyping*; Social networks; **Activity centered design**; • **Networks** → *Network security*; Transport protocols; Routing protocols; • **Security and privacy** → *Human and societal aspects of security and privacy*; • **Information systems** → *Web interfaces*; • **Applied computing** → *Media arts*;

Keywords

Alternate Reality Game; Cybersecurity; Zero-Dollar Budget; Player Prototyping

1 Related Works

Cybersecurity-themed alternate reality games are relatively new, yet are fast-growing and have a steadily increasing interest level in today's culture. In 2021, Giboney et. al [1] presented research on Cybermatics, a five-day "playable case study" inspired by ARGs where students spend five days in a virtual environment playing

the role of a cybersecurity penetration tester. Throughout the game, players uncover a storyline about a rogue employee for a company who has gained unauthorized access to the company's systems.

In addition, Tharot et. al [2] identified a widespread shortage of realistic cybersecurity game design elements available. The researchers developed a training game involving both attacker (red) and defensive (blue) elements while incorporating Microsoft's STRIDE methodology (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Escalation of Privilege [3]).

At the Polytechnic University of Turin in Italy, Lupano et. al [4] developed an ARG named OPUS, which was designed to teach students the concepts of securing an SQL database by exploring a cover-up by a fictional multinational organization. Both OPUS and Find Alex were marketed primarily on a college campus with QR code flyers. They are primarily web-based, utilizing fictional websites and social media pages for characters and organizations.

Similarly, Gestwicki and Stumbaugh [5] conducted a comparison analysis of 21 cybersecurity-themed games, classifying them into three types based on the extent to which game mechanics and player decision-making affect learning outcomes. The authors found a significant lack of cybersecurity-themed games that fit a perfect blend of accessibility for new players and an accurate portrayal of cybersecurity concepts.

2 Introduction

Alternate reality games (ARGs) are transmedia experiences that typically require 3 months or more to play, often defining themselves as a 'this is not a game' experience. Players of ARGs often navigate through physical or digital spaces, attempting to solve puzzles within a realistic setting to uncover a fictional narrative.[6] Find Alex is an asynchronous ARG developed at the University of Central Florida (UCF) by graduate students in the Digital Media program. The developers were 15 students divided into domain-specific teams for narrative, design, art, marketing, and mechanics. An informative video detailing many features of Find Alex can be accessed here: https://drive.google.com/drive/folders/1SYnuJ3c91sjGsPicRdIK0dJ67fhg01T1?usp=drive_link.

The Find Alex developers created an experience for players that exposes them to cybersecurity concepts and enables them to play as both a cyber-attacker and a defensive expert within the context of a mystery game. The work was developed as affordably as possible, using social media platforms such as Instagram [7] and Reddit [8] while employing inexpensive solutions such as automated emails and zero cost web hosting. This approach ensures the game includes

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diverse content and multimedia without additional development costs. To maintain the excitement found in synchronous ARGs, mechanics were designed to allow for optional hint utilization and player-to-player communication.

3 Find Alex Technical Overview

Find Alex consists of websites and social media accounts designed to be visited sequentially. Each site or account provides new information to uncover about the characters, and each site has embedded puzzles to allow players to progress. Each website also has a unique Google tag for trackable Google Analytics metrics [9], in addition to seven checkpoints where developers can log player progression: tracking interactions, time spent on pages, clicks, and navigation. The technical overview presents the chronological order and design considerations that players experience across these sites.

4 Find Alex Page

Players of the game who find the "rabbit hole" poster with the QR code will be redirected to a website at <https://findalex.netlify.app/>.

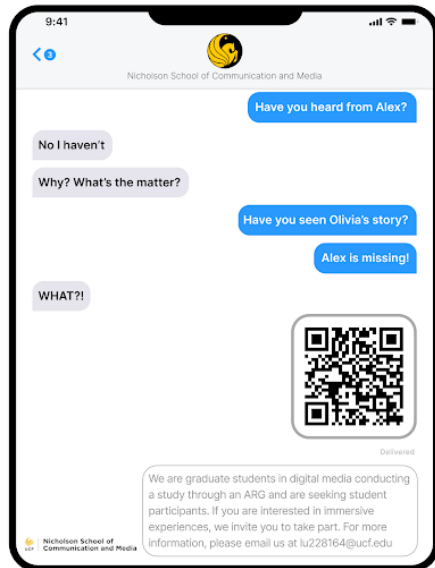


Figure 1: Find Alex QR Code Poster

This website serves as the entry point into the ARG and describes a missing person (Alex Valegro) and subsequent efforts by the family to spread news. The Find Alex page was designed to appear created by Olivia Holtz, a fictional in-game character who communicates with the player via email later in the storyline. The page presents information developed by the narrative team, describes Alex, and lists external links to Alex's Instagram, Wix Blog [10], and Olivia's Reddit (<https://www.reddit.com/user/Oliviaholtz/>). To facilitate a sense of community partnership in finding Alex, comments on this page are displayed from made-up users attempting to gain access to a locked Google Drive, foreshadowing the player's experience. Players are then prompted to enter their email to sign up for the search. The output of the sign-up process is an email from Olivia, which directs players to explore more of Alex's Instagram (<https://www.instagram.com/alex.valegro/?hl=en>) and Wix blog (<https://alexvalegro.wixsite.com/alexvalegro>) profiles. Find Alex is hosted on Netlify [11], a popular free static-site platform, to give the ARG's website an authentic, DIY feel, as if it were built by one of Alex's friends.

[//alexvalegro.wixsite.com/alexvalegro](https://alexvalegro.wixsite.com/alexvalegro)) profiles. Find Alex is hosted on Netlify [11], a popular free static-site platform, to give the ARG's website an authentic, DIY feel, as if it were built by one of Alex's friends.



Figure 2: Find Alex Landing Page

4.1 Locating the Google Drive

Within the Find Alex page, a blog post at the bottom from Olivia reveals a custom website link left by Alex that is meant to mimic an HTTP Error 403 page (<https://72211713181284141.github.io/HTTP-ERROR-403/>). After refreshing the page, players are prompted to enter their email and a password; to proceed, players must back-track to discover the hidden password and email within images on Alex's Instagram or personal blog. Alex's blog was included to allow players without an Instagram account to access her email and the required password hidden in the images. Using this information, Alex's Google Drive is unlocked for viewing at <https://drive.google.com/drive/folders/1F5EMhdPu1QXwjTOi9sqS0aWHIUGMmgay>.

At this point, players receive Olivia's second email, encouraging them to search further in Alex's files. The second puzzle resides within Google Drive: by navigating through the drive folders and finding specific images and messages, players unlock each folder in order (Conventions > Photography > Coding) with a code hidden in the other folders. Once the zipped folders are unlocked, the final image in the "Coding" folder reveals the URL of a mysterious club—one deeply connected to Alex's story and critical to advancing the game's narrative.

4.2 Club Page Level 1

Players who successfully unlock the third (Coding) folder on the Google Drive are presented with the URL for the Draconis Squamae club page (<https://www.draconis-squamae.ct.ws/>), where they can explore what appears to be a straightforward university club website.

Level One of the Draconis Squamae site was designed to display the publicly available information a visitor would typically find on any standard academic club website. The user interface was modeled after several pre-existing club websites from the University

of Central Florida. Level One includes a Home page, About page, Events page, and options to contact or join the club, which involve further progression in the ARG.

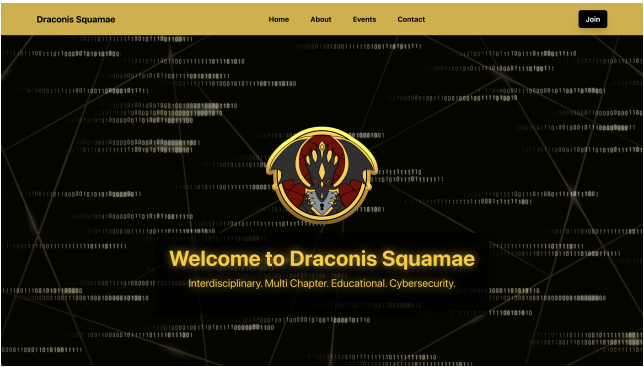


Figure 3: Draconis Squamae Club Home Page

Joining the club requires the player to enter a username into the join bar, another trackable location for the game developers to monitor overall player progression. Image verification is required after entering a username, which utilizes hash values for images, comparing the player's uploaded images (found in the Google Drive) against a standardized image stored locally on the site by the developers.

Once verified, the player gains access to a technical initiation test required to become a club member, presented as a command prompt-style terminal. To progress, players must use the "Terminal 1 Codes" list, labeled ImportantCalculator.html, in the Coding folder on Google Drive.

Once players determine the correct commands from the code list, they can copy and paste them into the terminal and progress through the initiation, where a hidden section of the website (Level 2) is revealed directly below the terminal. Terminal 1 was intended to teach players basic safe online browsing habits on the client side while slowly introducing programming languages. Players can use the club's virtual private network (VPN), enter Python-based commands, and import libraries in a directory for navigation.

4.3 Club Page Level 2

The second level of the club page is similar to the first, with additional information displaying key club members and information regarding Alex. However, players will notice that a "do not associate list" is present with additional information such as the club member IDs, leading the player to believe that the club is more abnormal outside the fictional context. To progress, players must find the word Trojan on the page, which will unlock the final locked Google Drive folder, Apple. Inside the Apple folder, players will notice that the documents all are related to Alex's last pieces of evidence which she gathered before disappearing, including a new codes list named "AlexNotes.html" where Alex noted an updated script that could be entered back into the routine access terminal to unlock new secrets about the club. Specifically, this script is based on elevating player access privileges to the sudo level, a commonly used permission for Unix/Linux, enabling users to run commands at the root level.

Upon entering this script, players will be redirected from terminal 1/club level 2 to a new page within the club site (level 3).

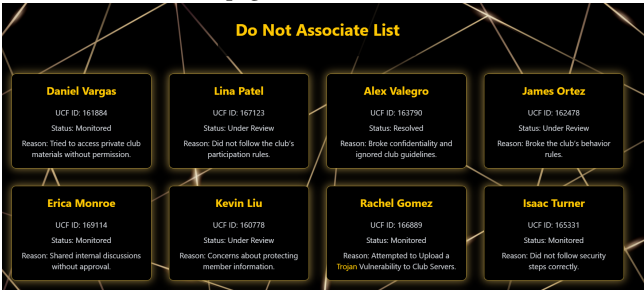


Figure 4: Draconis Squamae Club Level 2 Page

4.4 Club Page Level 3

Level 3 of the club page has a different, darker look with a corresponding level of more sinister club content, including reasons why members were banned and information on the extent of the club's other chapters. Inside level three, players will find terminal 2, a pop-up window where players must enter additional commands to break down the website's design by referring to Alex's notes inside the Apple folder. Terminal 2 is focused on building on the initial concepts from Terminal 1, but with a focus on safe networking techniques such as port security, firewall rules, and vulnerabilities, in addition to web-based considerations such as domains and name services.

Initially, players will log on with Alex's credentials and enter "Alex's last command" before her mysterious disappearance. Players will then follow Alex's plan to expose the club, accomplishing what she had never been capable of doing: uncover the club's hidden file share, accessible to certain trustworthy members. It is at this point where the player begins to transition into a cyber attacker, ultimately inducing a vulnerability into the website to expose a hidden URL (<https://dracoremove14414292.rf.gd/>) and also a final remote access codes list.

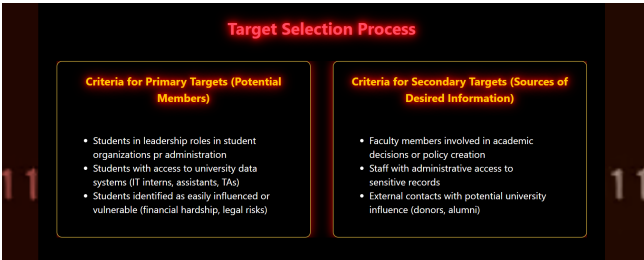


Figure 5: Draconis Squamae Club Level 3 Page

4.5 Club Page Level 4

Level four of the club page is themed to look like Tor Browser, a commonly accessed gateway into privately hosted web pages. In practice for the ARG, it hosts a terminal-themed interface which serves as the club's remote access file share, containing secretive and sensitive files, including audio and text-based blackmail, which the club keeps on certain members. To proceed, players return to

the remote access codes list exposed during the previous puzzle's completion. Terminal 3's codes list follows a similar methodology to the previous terminals in that it exposes players to more complex cybersecurity-themed concepts, mostly focusing on tunneling over networks and encryption. Players begin the terminal by logging in with the credentials uncovered by breaking into the club's website, and must subsequently run several scripts to progress.

A unique feature about level four is the end stage; once the player successfully breaks open folders with a rainbow table and runs a vulnerability script, they will be presented with an option to download files. Upon a successful download, publicly accessible information about the user, such as browser type and RAM, will be presented with a message informing the player that the data is being forwarded to club administrators.

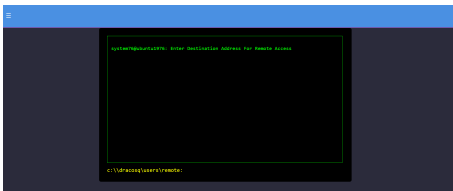


Figure 6: Draconis Squamae Club Level 4 Page

4.6 Final Upload Page

Players who return to the evidence upload URL provided by Olivia in her final email (<https://findalex.netlify.app/upload>) will be presented with a message requesting files to proceed. If the player uploads the required file from the previous file share, a series of mini games will begin with sound effects and animations on screen, giving the player a sense that the club is attempting to interfere with the file upload progression. Two mini games involve memorizing a pattern, while the others involve fast reflexes and button presses. Successful completion of all mini games will result in a completion screen, followed by an automated email to the player informing them of a newspaper article detailing the exposure of the club by law enforcement, and the final status of the investigation.

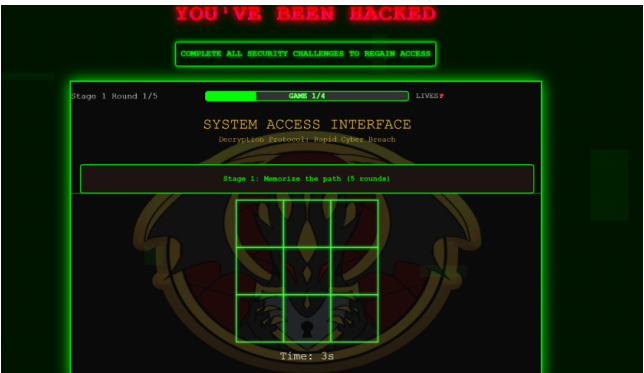


Figure 7: Final Evidence Upload Page

5 Working Within a Zero-Dollar Budget Environment

Find Alex was developed in a budget and time-constrained environment; the nature of the developers' status as full-time graduate students and the condensed developmental timeline of an academic semester dictated that a dedicated fundraising campaign was not an option. Developers of Find Alex employed a wide variety of zero-cost development, user interface design, and web deployment tools throughout the build process.

5.1 Zero Cost Coding Tools Utilities

A diverse selection of zero-cost coding, web utilities, and design enablers was used in the design process for Find Alex. Free social media platforms such as Instagram and Reddit were used in various capacities to serve as different stages of the ARG, where players needed to investigate posts and comments to proceed. Two Discord servers were created; one for developers and one for the player community to facilitate communication between design teams and player engagement. Google Drive [12] was also heavily used; to password protect folders on the drive, members of the mechanics team used 7-Zip's add to archive feature [13], which allowed password protection of folders using ZIP crypto. Once the folders were encrypted with passwords, they were zipped and uploaded to the live Google Drive site so that players would be forced to download, unzip, and enter the required password found throughout the various sections of the drive.

GitHub [14] proved to be a valuable resource, enabling the mechanics team to share the code for the websites and create various development branches. Branches were created before each new addition of content, which enabled a careful code review process to occur in a testing environment using a locally hosted server before pushing the changes to a live environment. Members of the mechanics team used the GitHub environment to create a repository for each of the dedicated websites, employing GitHub's student developer pack to expand the capabilities of each developed site.

Once the repositories were created, React [15] and Tailwind CSS[16] free utilities were employed to dynamically build each web page in a user-friendly way, which enabled increased creativity and more pleasant aesthetic decisions. One affordance of Tailwind is that it comes preset with a variety of add-ons designed to easily facilitate web content, such as the inclusion of image carousels, video content, or background images. React is a flexible web framework that allows creators to automate website page routing while providing a simple method to code multiple web pages and manage their interactions. In addition, simpler sites that had less content (the dead-end page and final terminal) were coded using basic HTML and plain text to allow for versatile deployment.

5.2 Zero Cost Management Tools

The developers of Find Alex also sought to utilize a plethora of zero-cost management tools to deploy the websites and analyze user traffic. Each site was deployed with zero costs; two were deployed using GitHub Pages, two were hosted on Infinity Free [17], one was deployed using Netlify, and one was created using the Wix environment. Two websites involved the creation of customized domain names on the ct-ws and rf-gd domains, employing free

SSL certificates from Google Trust [18]. To manage these two customized domains, an Amazon Web Services [19] free cloud plan was created using a zero-cost budget to maintain daily visits to the Infinity Free sites for longevity of play.

One critical tool heavily used for automated email communication to the players was Google Apps Script [20], a scripting environment designed for developers to work with Google Workspace [21]. This allowed for a seamless integration of Gmail [22], Sheets [23], and Google Drive, ultimately providing a custom and cost-effective solution for the various dynamic needs of the ARG. Sheets were used as a primary database for storing and managing players' data. It kept track of the player's emails, the current stage the player was at in the game, conversation count, and timestamps of interaction. The Sheets system was able to update players' progress in real-time and handle the necessary functions to determine which automated emails to send out or use in response to players. Additionally, the Sheets environment allowed for quick and easy data modification and ensured data would be maintained without external database services.

Email served as the communication foundation of the ARG, with automated emails being sent out to players at various stages of the game. It begins at the first web page players visit, where they are prompted to join the search for Alex by entering their email, triggering Olivia's first automated email response. This also updates the player's stage in Sheets to "registered", initiating the tracking system across websites. The other emails are triggered at key points throughout the game to deliver a layered narrative flow, facilitating Olivia's involvement in both game mechanics and overall story. When the player unlocks Alex's Google Drive, they can respond to Olivia's email with the code to proceed to the game stage status. This approach supported a carefully tailored response system between the players and Olivia, which presented developers the ability to add personalization, lead player progress, and enhance the overall immersion of the game.

Google Drive was also deployed to host related images used in the custom emails that could be easily fetched through Apps Script. The tool created a cohesive digital ecosystem within the Google Workspace to add integral functionality to the ARG while eliminating costs and improving development.

6 Prototyping for Players

In later stages of the development process, a formal playtest was scheduled with approximately 20 new players. Playtesting outputs included additional features and changes to various elements in the ARG's structure, appealing to the player styles exhibited during the playtest. In addition, adjustments and refinements were made during early access to make the experience enjoyable while keeping players from directly contacting the developers for help.

One of these early access discoveries was that, technically, component players were finding the public GitHub repositories for the websites where the code was stored and were emailing the design team directly, breaking player immersion. To resolve this issue, personal developer information was sanitized from GitHub, and the websites used by players were pointed in a different direction to ensure that players would not have any knowledge of the repositories. Another play tester revealed that the password for websites such

as the locked Google Drive could be easily found using developer tools to examine the website code. This was also problematic on the image verification login, which had strings of image names the player could easily find within the code, bypassing routine puzzle solving. To solve this issue, Mechanics made a concerted effort on every website to disable developer shortcuts on Mac/PC, disable right click, and obfuscate/hide obvious code to make the passwords and image names more difficult to find. This decision was intended to allow the storytelling of cybersecurity to still prevail for tech-savvy players by making it possible, yet more challenging for them to access the required information in the code while removing a shortcut that can bypass intended puzzles. These adjustments to the ARG were largely managed by a comprehensive flowchart, which tracked the game flow and changes, and was shared amongst the various design teams as seen below.

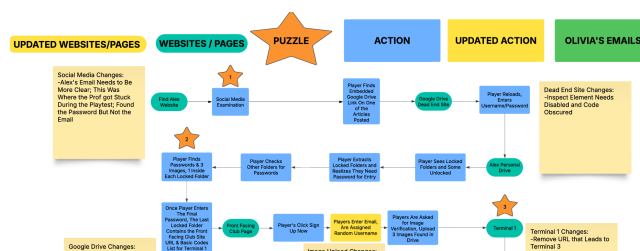


Figure 8: Puzzle Flowchart Used By All Teams Collaboratively

In addition to catering to technologically experienced players, play testing revealed that some parts were too convoluted for players. One example was the login verification puzzle, which asked for three images found in older parts of the Google Drive. After determining that all images needed similar motifs, mechanics further improved this puzzle by creating visual indicators that appeared for correct and incorrect images, which built confidence within players by keeping them on the right track. Image verification was additionally limited to only 3 images to prevent players from over-scoping the puzzle, and a message was added on the web page to ensure that players knew to only upload 3 images.

Hints are often provided in escape rooms or ARGs to help players redirect to the right path, and specifically, well-timed hints are critical in developing an effective puzzle [24]. The primary objective of the hint strategy within Find Alex was to avoid providing too many hints or obvious answers while at the same time ensuring that sufficient detail was included so players knew how to proceed. Therefore, the hint system in Find Alex serves more as guide rails to keep the player on track linearly, rather than providing specific context for a puzzle or game stage, which might be confusing.

To facilitate balancing, players were contacted directly through their email address, which they were encouraged to enter on the first screen. At each stage in the ARG, a specific action or button that the player interacts with will trigger an automated email from Olivia (Alex's friend), who will guide how to proceed in the player's journey. These emails are optional for the player to check, allowing players to complete the ARG without help. The Olivia emails are vague enough to point in the right direction without giving additional information on what should be looked for or what the

correct solution is. An example of this is in the first email, where Olivia implies the existence of a locked Google Drive folder, which has a prerequisite of finding Alex's email and password.

Finally, there was a need for additional player assistance, so the Find Alex community was created on Discord [25], a social networking platform that caters to gamers and content creators. Inside the community organization is a specific channel for player-to-player communication, which has a two-hour auto-delete timer to prevent spoilers from being archived. This feature enables players to ask context-specific questions to other players who may be further ahead in the ARG, and also facilitates a shared sense of collaboration and teamwork.

7 Realistic Cybersecurity Elements

Find Alex has a variety of realistic cybersecurity and network hardening concepts, some designed to be overtly obvious to the player, while others are covert. Developers of Find Alex included references to safe and secure cybersecurity strategies throughout the ARG, on the websites, inside the terminal codes lists, within the terminal puzzles, and within the developer strategy to preserve user data and limit intrusiveness. Capture the flag exercises, cyber games, and penetration testing tabletop events are commonly used by cybersecurity professionals and organizations; red and blue teams are typically used to organize employees during these events [26]. Blue teams are tasked with defending and hardening systems, while red teams are tasked with breaking in and intruding on systems. In addition, white teams are typically employed to ensure both sides meet the organization's objectives within the exercise parameters. The developers of Find Alex used this framework within a game environment to funnel players into blue and red team members throughout playing the ARG.

7.1 Decisions to Support Cybersecurity Themed Outcomes

In the early stages of ARG development, Find Alex developers worked to design the puzzles that intertwined a captivating cyber-themed narrative with challenging programming games to achieve the goal of educating players on cybersecurity. As players progress using terminals as a medium to complete the puzzles, the game's narrative unfolds procedurally as players first explore, then hack into various websites. One of the ARG's cyber-related themes involves challenging the player's initial perception of online security through efficiently dismantling secure elements such as site protections, firewalls, and network security settings.

In addition, Olivia's ability to track the player throughout the game, providing passive hints to the player through their email, is a subtle reminder of the player's own web insecurities and lack of understanding of how their data is being used. Within puzzles, there are prompts for accessing the player's microphone or camera; these are untracked actions, but intended to encourage players to reflect internally. For example, a player might consider how quickly they accept a user agreement or click yes to various online requests to expedite their goal. Finally, the utility of Alex's data comes into question when the player realizes that ultimately her data sharing, storage choices, and social media decisions all contributed to her disappearance.

Players who complete the ARG will inadvertently be challenged to examine their own browsing, file storage, and cybersecurity habits, particularly when their own system information appears on screen at the completion of terminal 3. Therefore, each design and narrative decision element of the ARG was implemented by the mechanics team to contribute wholeheartedly to the goal of cybersecurity awareness and education.

7.2 Player as a Blue Team Member

In the early stages of the game, the players employ defensive cybersecurity strategies and concepts when interacting with the terminals and websites. On the initial club page, players will find two training courses designed to mirror a COMPTIA Security+ and CEH (Certified Ethical Hacker) training course. Also embedded within the landing page is a link to "Hack the Box" [27], a hyper-realistic cybersecurity capture the flag exercise designed to train corporate cyber experts on hardware/software hardening and vulnerability protection.

As players progress and begin to enter commands into a command prompt-style terminal, they'll be referring to a code list required to proceed through each terminal. Embedded into the code lists are additional cybersecurity concepts, such as the value of using a virtual private network (VPN), proxy server, and secure shell (SSH) for remote connections. After reading through the code lists, players will connect to a VPN, employ SSH for remote access, and view port security. Developers of Find Alex carefully selected these actions as they teach valuable concepts for students unfamiliar with safe online browsing, cybersecurity and networking habits.

7.3 Player as a Red Team Member

As the game progresses, players will be exposed to known vulnerabilities, exploits, and techniques that attackers use to target websites and personal data. In the middle stages of the ARG, players will see that the club which they are investigating is involved in nefarious activities, so the players begin to use techniques employed by the club against the club itself to discover the true nature of the organization and the circumstances surrounding the disappearance of Alex. Some of the later codes lists reveal that the club is targeting specific members and teaching members how to inject vulnerabilities into systems. For example, players will discover that the club is showing members how to employ rainbow tables to break encrypted files, disabling CloudFlare [28] web protection to attack online sites, using Tor browser to host dark web blackmail, disabling port security, and utilizing the EternalBlue vulnerability.

As the player continues to dig, they'll eventually use all of these vulnerabilities against the club to expose the organization. The systematic process of breaking open encrypted files and damaging websites was a specific design choice intended to make the user question their habits and security while acting as a Red team member.

8 Future Work

While Find Alex was developed in a class environment with a set production and end planned, the development process offers insights for future projects, such as the zero-budget development

approach and the integration of free web hosting platforms streamlined with automated deployment processes. The DIY aesthetic achieved through zero-cost tools proved engaging and could be further leveraged as an intentional design choice rather than a limitation. Additionally, the combination of free analytics tools could be developed into a comprehensive player tracking framework, allowing developers with limited resources to create more responsive game experiences.

9 Conclusion

Find Alex is an excellent use case for a student-crafted project built with zero developmental costs, subscriptions, or tiered service restrictions. The tools described in this analysis are recommended for future alternate reality game designers with limited funds. The ARG is not intended to be a cybersecurity training system, nor intended as an online mystery game with no ties to realistic cyber attributes. Developers of Find Alex intended the ARG to be a narrative-driven storyline with puzzles and games rooted in exposing players to cyber and network security fundamentals while still maintaining a high level of player intrigue and enjoyment. Find Alex was developed with input from various domain-specific teams and refined through extensive playtesting and player feedback. This design process enabled us to fine-tune the player experience dynamically, rather than building a product inside a vacuum. In closing, Find Alex is a valuable example of how an inexpensively crafted asynchronous ARG can provide educational outcomes relevant to society while simultaneously providing an engaging experience for players.

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